

Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

ABDULRAHMAN ALSWAIDAN, Cornell University, USA

JEFFREY VARNER, Cornell University, USA

Generating synthetic financial time series that preserve the statistical properties of real market data is essential for stress testing, risk model validation, and scenario design, yet existing approaches, from parametric models to deep generative networks, struggle to simultaneously reproduce the heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering that characterize empirical equity data. We propose a hybrid hidden Markov framework that discretizes continuous excess growth rates into Laplace quantile-defined market states and augments the regime-switching process with a Poisson-driven jump-duration mechanism to enforce realistic tail-state dwell times. Parameters are estimated by direct transition counting, avoiding the Baum-Welch EM algorithm entirely. We evaluate synthetic data quality using three complementary metrics: Kolmogorov-Smirnov and Anderson-Darling goodness-of-fit pass rates for distributional fidelity, and ACF mean absolute error for temporal structure preservation. Applied to ten years of SPY data across 1,000 simulated paths, the framework achieves KS and AD pass rates exceeding 97% and 90% in-sample and 95% out-of-sample (full calendar year 2025), reproducing the ARCH effect that standard regime-switching models miss. GARCH(1,1), by contrast, achieves only 5.5% and 1.9% in-sample pass rates despite its well-known volatility clustering properties. A Single-Index Model extension propagates the SPY factor path to a 424-asset universe, enabling scalable correlated synthetic path generation while preserving cross-sectional correlation structure.

Additional Key Words and Phrases: Synthetic Data Quality, Time Series Generation, Hidden Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching, Stylized Facts, Distributional Fidelity

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance and data quality research. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [2, 24]. Yet the quality requirements for financial synthetic data are unusually demanding: empirical equity excess growth rates exhibit three well-documented statistical regularities, namely heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering, that any generative framework must simultaneously reproduce to be considered statistically faithful [4, 10, 30]. These domain-specific quality criteria, often called *stylized facts*, serve as the benchmark against which the fidelity of synthetic financial data must be evaluated.

Existing generative approaches address these quality requirements from different angles but each falls short on at least one dimension. GARCH-family models [8, 13] capture conditional variance dynamics but do not explicitly represent discrete market regimes or sudden jumps in volatility. Stochastic-volatility and jump-diffusion models [22, 32] enrich tail behavior but lack a mechanism for volatility persistence after jumps. Deep generative models, including recurrent neural networks [16, 23] and generative adversarial networks [29, 42, 43], can learn complex distributions from data but struggle to reproduce temporal dependence structures, particularly volatility clustering,

Authors' Contact Information: Abdulrahman Alswaidan, Cornell University, Robert Frederick Smith School of Chemical and Biomolecular Engineering, Ithaca, NY, USA; Jeffrey Varner, Cornell University, Robert Frederick Smith School of Chemical and Biomolecular Engineering, Ithaca, NY, USA, jdv27@cornell.edu.

unless the network architecture is specifically designed to capture them [4]. Hidden Markov models (HMMs) and related regime-switching frameworks offer a probabilistic structure for non-stationary market behavior through latent state dynamics [35], yet standard HMM specifications fail to generate persistent high-volatility regimes, leading to overly rapid reversion following extreme events and poor temporal quality in the synthetic output [9, 37]. Bridging this gap requires a generative framework that preserves both distributional fidelity and temporal structure while remaining computationally scalable.

We address this challenge by developing a hybrid HMM framework that discretizes continuous excess growth rates into quantile-based regimes and augments the resulting Markov process with a two-parameter Poisson jump-duration mechanism that forces the model to dwell in high-volatility tail states for empirically realistic durations, connecting gradual price evolution with sudden large moves identified by jump-detection methods [3]. By partitioning excess growth rates into quantile-defined states via a Laplace cumulative distribution function, we enable direct frequentist counting of regime transitions, bypassing the iterative Baum-Welch algorithm (and its sensitivity to starting values) entirely, making the framework scalable to a 424-asset synthetic data pipeline. We evaluate synthetic data quality using three complementary metrics: Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) pass rates for distributional fidelity and autocorrelation function mean absolute error for temporal structure preservation, all reported with standard errors. Using ten years of SPY data (2014–2024) for training and 249 trading days (full calendar year 2025) for out-of-sample testing, we demonstrate KS and AD pass rates of 97.0% and 90.5% in-sample and 95.0% and 95.9% out-of-sample across 1,000 simulated paths, with volatility clustering preserved in both windows. A Single-Index Model factor extension [39] propagates the univariate SPY engine to the full 424-asset universe in a single generative pass, preserving cross-sectional correlation structure. We find that the jump-duration extension is necessary for reproducing the persistent high-volatility regimes that standard regime-switching models miss.

2 Related Work

2.1 Stylized facts and the limitations of parametric models

The quantitative study of financial market dynamics has been shaped by a tension between tractable theoretical frameworks and the empirical complexity of observed data. The Black-Scholes-Merton option pricing model [7] and the geometric Brownian motion (GBM) paradigm embedded within it assume that log-returns are independently and identically distributed Gaussian random variables, producing asset prices that move smoothly (no sudden jumps) with fixed-size random fluctuations and thin tails that assign negligible probability to extreme moves. These properties carry well-known practical advantages: closed-form option prices, simple portfolio variance calculations, and minimal data requirements. However, the Gaussian diffusion paradigm conflicts with a broad set of empirical regularities documented well before the model’s widespread adoption. Mandelbrot [30] demonstrated that speculative price changes exhibit heavy tails far beyond Gaussian predictions, a phenomenon now described as *leptokurtosis*. Fama [14] showed that raw return series display negligible linear autocorrelation, consistent with the efficient markets hypothesis. Yet Schwert [38] and others documented that the *magnitude* of returns exhibits substantial positive autocorrelation decaying slowly over time: large price changes tend to be followed by large changes regardless of sign, a pattern referred to as *volatility clustering*. Cont [10] provides a systematic review confirming these regularities across asset classes, geographic markets, and sampling frequencies. Collectively, leptokurtosis, the absence of return predictability, and volatility clustering define the benchmark against which any model of market dynamics must be evaluated.

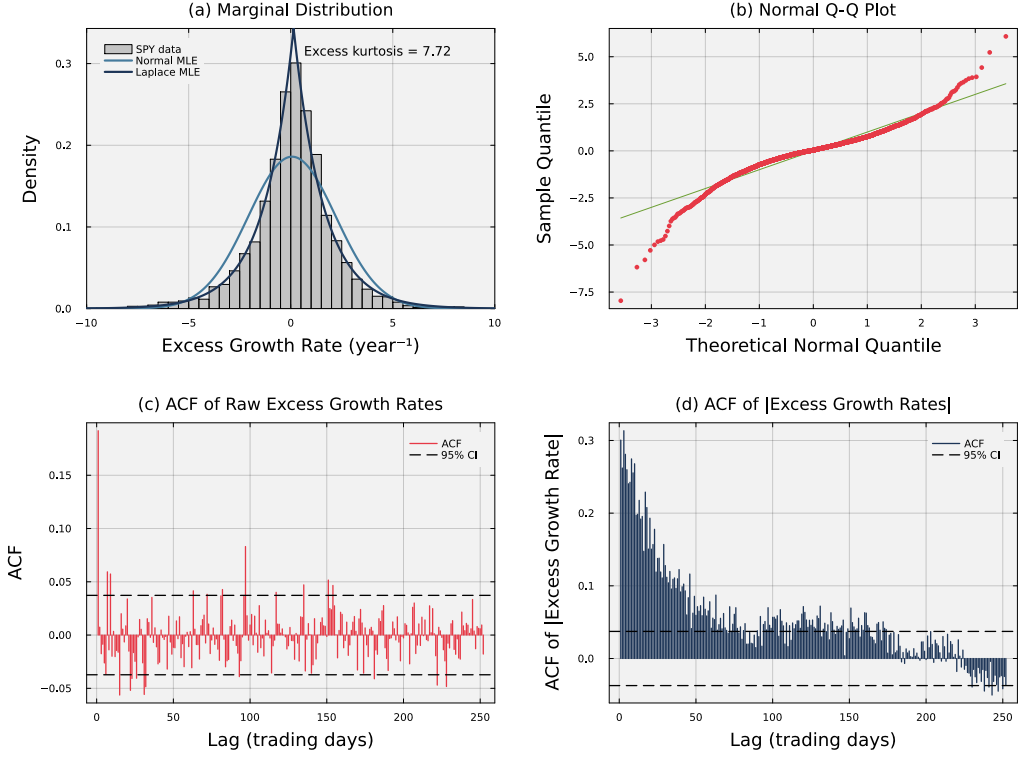


Fig. 1. Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns (consistent with the efficient markets hypothesis) against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

The most influential class of models designed to address volatility clustering within the Gaussian framework is the autoregressive conditional heteroscedasticity (ARCH) family introduced by Engle [13], generalized by Bollerslev [8] to the GARCH(p,q) specification in which the conditional variance depends on both past squared residuals and past conditional variances. GARCH models are parsimonious and widely used, but they have well-documented limitations. Heavy tails must be injected by choosing a fat-tailed noise distribution (e.g., Student-t); the model itself does not produce them. There is no representation of discrete market regimes or sudden jumps. Volatility persistence is controlled by a single decay parameter, so the model cannot distinguish between a calm market that slowly heats up and a sudden crash that lingers [12]. Stochastic volatility models [22, 41] model volatility as a hidden quantity that evolves randomly over time, addressing some of these shortcomings, but require computationally intensive estimation and their parameters do not directly correspond to observable market states. An alternative approach to modeling fat tails and discontinuous price behavior is the explicit introduction of Poisson-driven jumps into the asset price process. Merton [32] added a compound Poisson jump component to GBM, producing a Gaussian-mixture marginal distribution that directly addresses leptokurtosis, and Kou [28] extended this framework with asymmetric jump sizes to capture the empirical pattern that crashes tend to

be sharper than rallies. Jump-diffusion models are particularly relevant in equity markets, where macroeconomic announcements, earnings surprises, and geopolitical events generate discrete large-magnitude price movements [3]. However, in their standard form, jump-diffusion models do not provide a mechanism for volatility persistence: the post-jump volatility immediately returns to its baseline level, absent a separate stochastic volatility component.

2.2 Hidden Markov models and regime-switching

Hidden Markov models (HMMs) provide a probabilistic framework for non-stationary time series in which the observed data are generated by a process that switches among a discrete set of latent states according to Markov dynamics. The foundational algorithmic framework was developed by Rabiner and Juang [35], who introduced the Baum-Welch algorithm for maximum-likelihood estimation of emission and transition parameters from observed sequences. Hamilton [19] applied the regime-switching HMM to macroeconomic time series, demonstrating that U.S. GDP growth is well described by a two-state model with distinct expansion and recession regimes, establishing the template for latent-state modeling of financial time series. The identification of time-varying market regimes has since become a central application of HMMs in finance, with uses ranging from stock return modeling and volatility forecasting [36] to tactical asset allocation [33].

Rydén, Teräsvirta, and Åsbrink [37] provided an influential systematic evaluation of HMM specifications against the stylized facts of daily return series, showing that simple Gaussian-emission HMMs with a small number of states could reproduce leptokurtosis and the absence of return autocorrelation, but failed to generate the observed volatility clustering; the ACF of absolute returns under standard HMMs decayed far too rapidly relative to empirical data. This diagnostic limitation identified the core structural gap that the present study addresses. Bulla and Bulla [9] advanced this line of inquiry by considering hidden semi-Markov models, in which the time spent in each state can follow a flexible distribution rather than the simple memoryless (geometric) decay of standard Markov chains, and showed that this extension substantially improves the reproduction of volatility clustering by allowing the model to remain in high-volatility states for realistic durations. A persistent practical limitation of both standard and semi-Markov HMM approaches is the computational burden of maximum-likelihood estimation via the EM algorithm, particularly when the number of states is large or the asset universe is broad: the Baum-Welch procedure requires iterative forward-backward passes and may converge to local maxima depending on initialization. The present study addresses this limitation by replacing EM with direct frequentist counting of transitions between quantile-defined states, an approach that is computationally trivial and free of initialization sensitivity.

2.3 Synthetic data generation and deep generative models

The generation of synthetic financial time series has attracted increasing attention both as a practical tool for augmenting limited data, testing risk models, and stress testing portfolios, and as a benchmark for evaluating generative model quality. Assefa et al. [2] survey the opportunities, challenges, and pitfalls of synthetic data generation in finance, noting that stylized-facts reproduction is the primary criterion for evaluating statistical quality and that naive generative approaches tend to fail on at least one of the three core stylized facts. The rise of generative adversarial networks (GANs) prompted several investigations into whether deep learning architectures could learn these properties from data. Takahashi, Chen, and Tanaka-Ishii [42] showed that standard GAN architectures produce synthetic return sequences that visually resemble financial time series but fail to reproduce ACF structure precisely; Kwon and Lee [29] provided a careful evaluation against the stylized-facts checklist, finding that while GANs match marginal distributions reasonably well, capturing temporal dependence structures, particularly volatility clustering, remains challenging

unless the architecture is specifically designed to learn them. Foundation models for Markov jump processes [5] offer a complementary perspective in which large-scale pretrained inference networks can be applied to new process instances without retraining, but their application to financial time series generation remains an open research direction.

Against this backdrop, the hybrid HMM framework of the present study occupies a distinct position: it is explicitly designed around the structural properties known to generate stylized facts, uses an interpretable discrete-state representation that can be audited and stress-tested, and achieves strong statistical fidelity with minimal computational overhead. The empirical estimation procedure avoids the training instability and data requirements of deep generative models, while the jump-duration mechanism provides a principled solution to the volatility-clustering limitation of standard HMMs. These properties make the framework particularly well-suited for risk management and stress-testing pipelines, where interpretability and computational reproducibility are as important as statistical fidelity.

2.4 Factor models and multi-asset extensions

Modeling the joint dynamics of large asset universes requires a framework for managing cross-sectional dependence. The Single-Index Model (SIM) of Sharpe [39] provides the canonical linear factor decomposition: each asset's excess return is decomposed into a common market factor component and an idiosyncratic residual, drastically reducing the number of parameters needed compared with a full multivariate model. The SIM remains foundational in portfolio construction and risk management despite the availability of richer multi-factor models such as Fama-French [15], because its simplicity enables transparent attribution and scalable computation. Existing HMM-based work in finance has been predominantly univariate, focusing on single-asset or single-index regime identification. Applications to individual equities [33], volatility modeling [36], and macro time series [19, 25] collectively demonstrate the utility of latent state representations but leave open the question of how HMM-derived generative models can be extended to large cross-sections of assets in a computationally tractable way. Mixture hidden Markov models [11] offer a partial solution by allowing multiple assets to share a common latent state structure, but estimating such models jointly across hundreds of assets is computationally demanding. The present study addresses this gap by combining the univariate HMM framework with the SIM factor decomposition: the hybrid HMM is estimated on the SPY index, which serves as the market-factor generator, and asset-level paths are reconstructed via the linear SIM projection. This construction exploits the fact that most of the shared volatility clustering across equities is driven by the broad market, yielding a scalable generative pipeline for a 424-asset universe that inherits the stylized-facts fidelity of the HMM without requiring joint estimation of a high-dimensional model.

3 Methods

Data.

The empirical foundation of this study was a dataset composed of 424 United States-listed equities and exchange-traded funds spanning 10 years (2014-2024). A pipeline was developed to support the automated construction of excess growth rate models for any ticker within this dataset, allowing for batch simulation and stress testing across market sectors. To validate the performance of this pipeline, we focused our analysis on the single-asset SPY, which tracked the Standard & Poor's (S&P) 500 index. The data included daily open, high, low, and close prices along with volume-weighted average price metrics (VWAP). For training purposes, the first 2,766 observations constituted the in-sample dataset, spanning from January 3, 2014, to December 31, 2024. A subsequent 249 trading days, from January 2 to December 31, 2025, were reserved for out-of-sample testing to assess model generalizability.

Excess growth rate calculation.

We computed the excess growth rate $G_{i,j}$ for ticker i between time period $j - 1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share (USD/share)). To measure how much each stock's growth exceeded a risk-free baseline, we defined the excess growth rate as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

where Δt represented the time step set to $1/252$ for daily data (units: years), and r_f denoted the constant continuously compounded risk-free rate (units: year⁻¹) derived from Separate Trading of Registered Interest and Principal of Securities (STRIPS) bond yields observed on September 10, 2025. Unlike a simple return, the excess growth rate has units of inverse time (year⁻¹) and directly quantifies the premium earned above a continuously compounded risk-free benchmark; log growth rates are also time-additive, making them consistent with a continuous compounding framework. While r_f varied dynamically over the sample period, we used a constant proxy to isolate the volatility patterns in the data without adding noise from interest rate movements.

Discrete hidden Markov model.

We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [35]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood regimes and was defined by quantile bins of a fitted cumulative distribution function (CDF) for the excess growth rate series. The observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the stationary distribution. The state-conditional observation model specified that the excess growth rate, given hidden state k , followed a Normal distribution:

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \quad (2)$$

To map observations to hidden states during estimation, we defined quantile boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, with $Q_0 = -\infty$ and $Q_N = +\infty$ so that the partition covers the full support of the distribution. An observation was assigned to state k when $Q_{k-1} < G_t \leq Q_k$.

The choice of the Laplace distribution over alternatives such as the Student's t-distribution was motivated by its alignment with the peaks and heavy tails of high-frequency financial excess growth rates [10, 27]. The Student's t-distribution is often used to model leptokurtosis, but the Laplace distribution's sharper peak at the mean better matched the concentration of small price movements observed in the SPY dataset [21, 30]. The Laplace distribution also provided a closed-form expression for quantile estimation, whereas MLE for the Student's t-distribution required

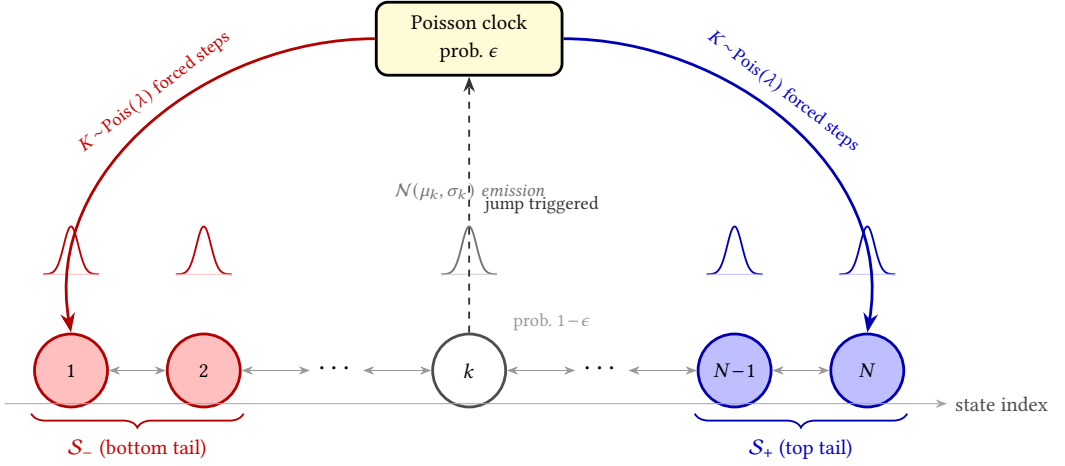


Fig. 2. Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix T (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ , dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set S_- (red, lowest-valued states) or the top tail set S_+ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional $\mathcal{N}(\mu_k, \sigma_k)$ emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

iterative numerical optimization [6]; this computational simplicity supported the 424-ticker pipeline. The quantile boundaries were derived from a parametric Laplace fit and therefore did not enforce equal counts per bin in the historical data; instead, they provided a partition of the distribution that preserved tail shape. By using the Laplace distribution to define the bins, the model established a baseline characterization of the return distribution's shape before the transition dynamics and jump mechanism were layered on top [28].

We estimated the transition matrix T through direct frequentist counting rather than iterative optimization. This approach differed from the traditional expectation-maximization (EM) algorithm used for parameter estimation in Gaussian mixture and hidden Markov models [6]. However, the discrete state assignments allowed for straightforward counting of observed transitions between regimes, making the approach both conceptually simple and free of initialisation sensitivity. A consequence of direct counting was that empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [9]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $S_{\text{bottom}} = \{1, \dots, N_{\text{tail}}\}$ and $S_{\text{top}} = \{N - N_{\text{tail}} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [17]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$,

where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k , the continuous excess growth rate ($\hat{G}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model captured sudden large price moves while keeping smooth variation within each regime [28]. The fitted Laplace CDF closely matched the empirical CDF for SPY, and the empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence (Figure S1, Online Appendix S1). Critically, under pure Markovian dynamics the model exited extreme (tail) states after only 1–2 steps on average, far shorter than the multi-week volatile episodes observed in real markets. This rapid reversion was the core limitation that motivated the jump-duration extension.

Growth rate generation.

The basic sampling procedure for the HMM (without jumps) proceeded as follows. The stationary distribution $\bar{\pi}$, representing the long-run probability of occupying each regime, satisfies the balance equation $\bar{\pi} = \bar{\pi}T$ and was estimated numerically by raising the transition matrix to a large power (T^{50}) [18, 20]. An initial hidden state was drawn from $X_0 \sim \bar{\pi}$, then for each subsequent step $n = 1, \dots, T$: (i) the next hidden state was sampled from the transition row $X_n \sim T_{X_{n-1}, \cdot}$, and (ii) an observation was sampled from the emission distribution $O_n \sim E_{X_n}$. This produced a sequence of hidden states $\{X_1, \dots, X_T\}$ and corresponding observations $\{O_1, \dots, O_T\}$.

Hyperparameter optimization via grid search.

To identify the optimal jump probability (ϵ) and mean jump duration (λ) [28, 32], we implemented a multi-objective grid search that minimized the discrepancy between historical and simulated market signatures. The error function targeted two stylized facts [10, 37]: volatility clustering, measured as the sum of squared errors between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [38] up to a lag of $L = 252$ trading days (approximately one trading year); and leptokurtosis, captured by a penalty term on the global kurtosis [30].

We defined the objective function $J(\epsilon, \lambda)$ as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L \left(\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau) \right)^2 + w_K \left(K_{obs} - \bar{K}_{sim} \right)^2 \quad (3)$$

where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth autocorrelation at lag τ and the global kurtosis, respectively. The simulated equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\bar{K}_{sim}$, were averaged across 200 independent synthetic paths of 2,766 trading days per grid point [17]. We set the kurtosis penalty weight to $w_K = 0.20$ to balance tail behavior against the temporal ACF term. The grid search explored ϵ values from 10^{-4} to 2.5×10^{-2} and λ values from 10 to 160. The full computational procedure is detailed in Algorithm 4.

Statistical validation metrics

To assess whether synthetic paths were statistically consistent with empirical data, we applied two nonparametric goodness-of-fit tests to the marginal distributions of simulated excess growth rate sequences. For each of the 1,000 simulated paths, the empirical cumulative distribution function (CDF) of the simulated sequence was compared against the empirical CDF of the historical in-sample or out-of-sample observations.

The Kolmogorov-Smirnov (KS) statistic is defined as:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \quad (4)$$

where F_n and F_m denote the empirical CDFs of the observed and simulated sequences of lengths n and m , respectively [26, 40]. The null hypothesis of distributional equivalence is rejected at significance level α when D_{KS} exceeds the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

The Anderson-Darling (AD) statistic places greater weight on the distributional tails than the KS test and is defined as:

$$A^2 = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln F(X_{(i)}) + \ln (1 - F(X_{(n+1-i)}))] \quad (5)$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics of the pooled sample and F is the empirical CDF of the reference distribution [1]. The AD test is particularly suited for assessing tail-distributional fidelity, which is the primary concern for risk management applications.

We reported the *pass rate* as the proportion of the 1,000 simulated paths for which the null hypothesis of distributional equivalence was not rejected at the $\alpha = 0.05$ significance level. A pass rate above 95 percent indicated that the synthetic paths were, in aggregate, statistically indistinguishable from the historical data under that test. Pass rates were reported separately for the in-sample and out-of-sample windows and for both model variants (HMM without jumps and the hybrid HMM with jumps).

Temporal fidelity metric

While KS and AD tests assess marginal distributional quality, they are insensitive to temporal dependence. To measure how well a generator reproduced volatility clustering, we defined the autocorrelation mean absolute error (ACF-MAE) as:

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \hat{\rho}_{|G|}^{\text{sim}}(\tau) \right| \quad (6)$$

where $\hat{\rho}_{|G|}^{\text{obs}}(\tau)$ and $\hat{\rho}_{|G|}^{\text{sim}}(\tau)$ are the sample autocorrelations of the absolute excess growth rate series $|G_t|$ at lag τ for the observed and simulated paths, respectively, and $L = 252$ (one trading year). An i.i.d. generator produces $\hat{\rho}_{|G|}^{\text{sim}}(\tau) \approx 0$ for all $\tau > 0$, so its ACF-MAE equals the mean level of the observed autocorrelation function; lower values indicate better reproduction of empirical volatility persistence.

Single-Index Factor Model extension

The framework described above applied to any individual asset ticker within the 424-asset dataset. To generalize synthetic path generation to a correlated multi-asset setting without fitting a separate HMM for each asset, we exploited the linear factor structure of the Single-Index Model (SIM) [39]. The SIM expresses the excess growth rate of asset i at time t as:

$$G_{i,t} = \alpha_i + \beta_i \cdot G_{\text{SPY},t} + \eta_{i,t}, \quad \eta_{i,t} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\eta,i}^2) \quad (7)$$

where $G_{\text{SPY},t}$ denotes the SPY excess growth rate at time t , β_i measures the systematic sensitivity of asset i to the index, α_i is the idiosyncratic drift, and $\eta_{i,t}$ is a zero-mean idiosyncratic shock assumed independent across assets and time. The parameters $(\alpha_i, \beta_i, \sigma_{\eta,i}^2)$ were estimated via ordinary least squares on the in-sample time series for each asset i .

Synthetic multi-asset paths were generated in three steps. First, the hybrid HMM produced a single synthetic SPY excess growth rate path $\{\hat{G}_{\text{SPY},t}\}_{t=1}^T$ using the regime-switching and jump-duration mechanism described above. Second, for each asset i , an idiosyncratic shock sequence $\{\hat{\eta}_{i,t}\}$ was drawn independently from $\mathcal{N}(0, \hat{\sigma}_{\eta,i}^2)$. Third, the synthetic excess growth rate for asset i

Algorithm 1 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M

Ensure: Simulated state sequence $S_{1:M}$

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1: Define tail-states  $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$  and  $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$ .
2: Sample initial state  $S_1 \sim \text{Categorical}(\bar{\pi})$ .
3: Set  $counter \leftarrow 2$ .
4: while  $counter \leq M$  do
5:   Sample  $u \sim \text{Uniform}(0, 1)$ .
6:   if  $u < \epsilon$  then
7:     Sample jump-duration  $K \sim \text{Poisson}(\lambda)$ .
8:     for  $j = 1$  to  $K$  while  $counter \leq M$  do
9:       Sample  $w \sim \text{Uniform}(0, 1)$ .
10:      if  $w < p_{neg}$  then
11:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$  {Bias toward negative-tail events}
12:      else
13:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ 
14:      end if
15:       $counter \leftarrow counter + 1$ .
16:    end for
17:  else
18:    Sample  $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1},:})$ .
19:     $counter \leftarrow counter + 1$ .
20:  end if
21: end while
22: return  $S_{1:M}$ 

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was recovered as:

$$\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{SPY,t} + \hat{\eta}_{i,t}. \quad (8)$$

This construction preserved the cross-sectional correlation structure induced by the common factor while maintaining the regime-switching and jump-diffusion dynamics of the index. The resulting 424-dimensional synthetic path could be used for stress testing, portfolio risk assessment, and scenario generation across the full asset universe in a single generative pass, avoiding the curse of dimensionality that would accompany a full multivariate HMM.

Simulation algorithm

Algorithm 1 details the core jump-diffusion simulation procedure; the remaining pipeline stages (model construction, state decoding, and hyperparameter grid search) are provided in Online Appendix S4.

Table 1. Descriptive statistics and stylized-facts tests for SPY daily excess growth rates. In-sample: 2014–2024 ($T = 2,766$); out-of-sample: 2025 ($T = 249$). Gaussian and Laplace columns show MLE fits to the in-sample data. JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $ G_t $ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

4 Empirical Study

4.1 Descriptive statistics and stylized facts

Descriptive statistics for the SPY daily excess growth rate series confirmed that the three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 249$) windows (Table 1). Excess kurtosis (kurtosis minus 3, so that a Gaussian distribution has a value of zero) reached 7.7 (IS) and 6.9 (OoS), the Jarque-Bera normality test was rejected ($p < 0.001$) in both windows, and the Ljung-Box test on $|G_t|$ was likewise rejected ($p < 0.001$), confirming leptokurtosis, negligible raw-return autocorrelation, and persistent volatility clustering. Any faithful generative model must reproduce all three properties simultaneously.

4.2 Jump hyperparameter optimization and state resolution

A multi-objective grid search over $\epsilon \in [10^{-4}, 2.5 \times 10^{-2}]$ and $\lambda \in [10, 160]$, with 200 simulated paths per grid point, identified a well-defined interior minimum at $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$, indicating that the jump hyperparameters were identifiable and that the optimization had a clear, unique minimum (Figure S2, Online Appendix S2). At these optimal values, the ACF of $|G_t|$ closely tracked the observed SPY ACF across lags 1–252, confirming that the jump mechanism reproduced the empirical ARCH effect.

The primary analysis used $N = 100$ states, but a sensitivity study over $N \in \{30, 60, 90, 100, 150, 200\}$ showed that the framework was robust to the choice of state resolution across a wide range (Table T1, Online Appendix S3). KS and AD pass rates remained stable for both HMM-NJ ($\geq 98.9\%$ KS, $\geq 97.9\%$ AD) and HMM-WJ ($\geq 96.0\%$ KS, $\geq 88.6\%$ AD), while ACF-MAE for HMM-WJ improved slightly from 0.057 at $N = 30$ to 0.049 at $N = 200$. State-occupancy diagnostics confirmed that every state at $N \leq 200$ contained at least 7 observations, ensuring adequate statistical support per state under the Laplace equal-probability partition. At $N = 350$, however, certain states were never visited in the historical record, leaving zero transition probabilities that prevented the model from reaching all states, which established a practical upper bound on state resolution.

4.3 In-sample distributional and temporal quality

We generated 1,000 synthetic paths of 2,766 trading days from each of six generators and evaluated them against the observed SPY series using two-sample KS and AD tests at $\alpha = 0.05$, along with kurtosis matching, ACF fidelity, novelty, diversity, and quantile coverage (Table 2).

Baseline anchors. We included three non-structured baselines to bracket the quality landscape. Bootstrap resampling of the training data served as a non-parametric upper bound: we observed 100.0% KS and 99.7% AD pass rates with 100% quantile coverage, but this generator simply recycled observed values and could not produce genuinely new distributional structure (0 estimated parameters). The Gaussian i.i.d. baseline (MLE fit) achieved 0.0% KS and 0.0% AD pass rates, which confirmed that a symmetric thin-tailed model was categorically inadequate for heavy-tailed equity returns. The Laplace i.i.d. baseline fared better at 44.0% (SE 1.6) KS and 43.3% (SE 1.6) AD, which showed that matching the tail decay rate alone recovered roughly half the distributional quality; the remaining gap reflected the regime-dependent variation that a single Laplace distribution could not capture.

Distributional fidelity. We evaluated KS/AD pass rates and kurtosis matching for the three structured models (Table 2). GARCH(1,1) achieved IS KS and AD pass rates of only 5.5% (SE 0.7) and 1.9% (SE 0.4), which indicated that the GARCH marginal distribution was systematically inconsistent with the observed data despite its well-known volatility clustering properties. HMM-NJ achieved 99.4% (SE 0.2) and 98.6% (SE 0.4), and HMM-WJ achieved 97.0% (SE 0.5) and 90.5% (SE 0.9), which confirmed that the Laplace-mixture emission structure adequately captured the observed heavy-tailed distribution. Simulated excess kurtosis told a complementary story: the observed value was 7.7, GARCH overshoot at 8.2 (SE 0.37) with high variance across paths, while HMM-NJ and HMM-WJ produced 5.7 (SE 0.02) and 5.5 (SE 0.03) respectively. The HMM variants underestimated kurtosis because the equal-probability Laplace partition smoothed the most extreme tail quantiles, but they did so with negligible path-to-path variation, which suggested a stable generative mechanism. Quantile coverage reinforced these findings: Bootstrap, HMM-NJ, and HMM-WJ all achieved 100% coverage of the 1st–99th empirical percentiles, while Gaussian covered only 13.1%, Laplace 37.4%, and GARCH 29.3%, which indicated systematic tail misspecification in all three non-HMM parametric generators (Figure 3, panel c).

Temporal fidelity. We measured temporal quality using the mean absolute error of $ACF(|G_t|)$ over lags 1–252. Bootstrap, Gaussian, and Laplace all produced an ACF-MAE of 0.060, which was expected because i.i.d. draws destroy all temporal dependence by construction. GARCH achieved the lowest ACF-MAE (0.031, SE 0.001) because its variance equation was explicitly designed to match autocorrelation structure; however, this advantage came at the cost of distributional fidelity, a tradeoff that the quality metrics exposed directly. HMM-NJ scored 0.059 (SE <0.001), essentially indistinguishable from the i.i.d. floor, which confirmed that the standard transition matrix alone was structurally incapable of generating persistent volatility clustering (Figure 3, panel b). HMM-WJ scored 0.052 (SE <0.001), the only non-GARCH model to reduce ACF-MAE below the i.i.d. baseline. The improvement arose because approximately 24% of HMM-WJ paths contained at least one Poisson jump event that sustained $ACF(|G_t|)$ decay across all lags, while the remaining 76% behaved identically to HMM-NJ. Importantly, the jump frequency is continuously tunable: increasing ϵ produces more jump-containing paths and further reduces ACF-MAE, at the cost of degrading distributional fidelity. The grid search selected $\epsilon^* = 10^{-4}$ because this value minimized the joint objective over both temporal and distributional quality (Equation 3), meaning the 24% jump rate was the empirically optimal operating point for SPY, not a structural ceiling. The jump-duration mechanism was therefore the structural feature responsible for reproducing the ARCH effect, with its strength directly controlled by ϵ .

Novelty, diversity, and parsimony. We computed novelty (mean correlation distance to the observed series) and diversity (mean pairwise correlation distance among synthetic paths) to verify that the generators produced genuinely new, non-redundant scenarios. All six models scored

Table 2. In-sample and out-of-sample model comparison for SPY ($N = 100, 1,000$ simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. HMM-NJ: standard Hidden Markov Model without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Novelty: mean correlation distance $1 - |\rho|$ between each synthetic path and the observed series. Diversity: mean pairwise correlation distance among synthetic paths. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors in parentheses: binomial SE for pass rates, $\text{std}/\sqrt{n}$ for kurtosis and novelty, bootstrap SE ($B = 500$) for ACF-MAE, diversity, and coverage. Arrows indicate preferred direction: $\uparrow$ higher is better, $\downarrow$ lower is better, $\approx$ closer to observed is better.

Metric	Bootstrap	Gaussian	Laplace	GARCH(1,1)	HMM-NJ	HMM-WJ
<i>In-sample: 2,766 trading days (2014–2024)</i>						
KS pass rate (%) $\uparrow$	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	99.4 (0.2)	97.0 (0.5)
AD pass rate (%) $\uparrow$	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	98.6 (0.4)	90.5 (0.9)
Excess kurtosis (observed)				7.715		
Excess kurtosis (simulated) $\approx$	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	8.2 (0.37)	5.7 (0.02)	5.5 (0.03)
ACF-MAE $\downarrow$	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	0.031 (0.001)	0.059 (<0.001)	0.052 (<0.001)
Novelty $\uparrow$	0.985 (<0.001)	0.984 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.984 (<0.001)	0.984 (<0.001)
Diversity $\uparrow$	0.985 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.984 (<0.001)	0.984 (<0.001)
Coverage (%) $\uparrow$	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.5)	100.0 (<0.1)	100.0 (<0.1)
<i>Out-of-sample: 249 trading days (2025)</i>						
KS pass rate (%) $\uparrow$	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	96.5 (0.6)	95.0 (0.7)
AD pass rate (%) $\uparrow$	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	96.8 (0.6)	95.9 (0.6)
Excess kurtosis (observed)				6.867		
Excess kurtosis (simulated) $\approx$	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	5.0 (0.08)	4.9 (0.08)
ACF-MAE $\downarrow$	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	0.026 (<0.001)	0.041 (<0.001)	0.040 (<0.001)
Novelty $\uparrow$	0.949 (0.001)	0.950 (0.001)	0.952 (0.001)	0.951 (0.001)	0.948 (0.001)	0.949 (0.001)
Diversity $\uparrow$	0.950 (<0.001)	0.949 (<0.001)	0.949 (<0.001)	0.950 (<0.001)	0.948 (<0.001)	0.948 (<0.001)
Coverage (%) $\uparrow$	100.0 (<0.1)	36.4 (1.9)	74.7 (1.1)	96.0 (1.3)	100.0 (<0.1)	100.0 (<0.1)
Parameters estimated	0	2	2	3	2	4

between 0.984 and 0.985 on both metrics, which confirmed that no generator memorized the training data and that synthetic paths were mutually independent. This uniformity was expected: all generators produced stochastic paths that were uncorrelated with any specific historical realization. Finally, HMM-WJ achieved the best joint distributional and temporal quality among all parametric models while requiring only four estimated parameters (N , ϵ , λ , and the transition matrix P , where N is set by design), compared with three for GARCH and two each for the Gaussian and Laplace baselines.

4.4 Out-of-sample evaluation

We evaluated all six generators on a held-out window of 249 trading days (January 2 through December 31, 2025) using the same metrics as the in-sample analysis (Table 2). Because the OoS window was roughly one-ninth as long as the training set, all two-sample tests had lower statistical power; pass rates therefore rose mechanically across every model, and the meaningful comparison was relative ranking rather than absolute level.

Baselines and distributional fidelity. We observed that Bootstrap resampling achieved OoS KS and AD pass rates of 97.4% (SE 0.5) and 98.5% (SE 0.4), remaining near the ceiling. The Gaussian baseline rose from 0% IS to 62.2% (SE 1.5) KS and 46.3% (SE 1.6) AD, and Laplace from 44% IS to 88.0% (SE 1.0) KS and 92.9% (SE 0.8) AD; these improvements reflected the reduced test power, not

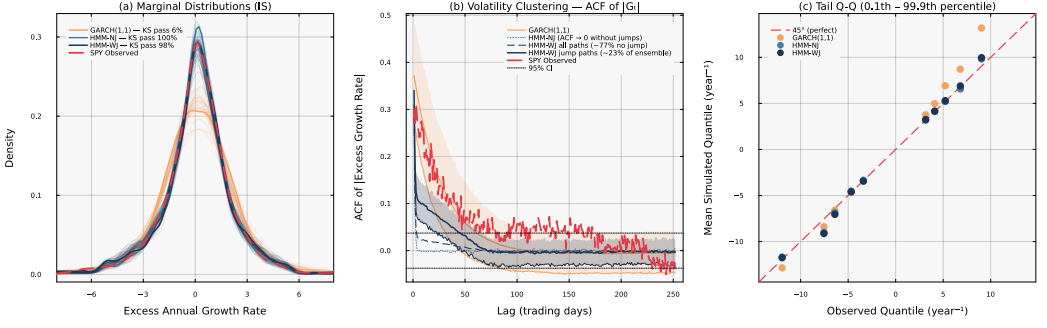


Fig. 3. Head-to-head in-sample model comparison for SPY ($N = 100, 1,000$ simulated paths). Panel (a): marginal density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 95% of paths (pass rate 5.5%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at rates $\geq 97\%$, confirming that the Laplace-mixture emission adequately captures the observed heavy tails. HMM-WJ generalises robustly out-of-sample (OoS KS pass rate 95%), while GARCH recovers partially (OoS 80%). Panel (b): autocorrelation function of absolute excess growth rates, $ACF(|G_t|)$, at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so $ACF(|G_t|) \approx 0$ for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining $\sim 24\%$ of paths contain at least one jump event and sustain substantial $ACF(|G_t|)$ decay across all lags (solid navy curve $\pm$ band). The jump frequency, and hence the fraction of the ensemble exhibiting slow ACF decay, is controlled by the hyperparameter λ , making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

genuine distributional fit, as confirmed by their kurtosis values (Gaussian 0.0, Laplace 2.7 against an observed 6.9). GARCH recovered to 80.3% (SE 1.3) KS and 72.0% (SE 1.4) AD, but its simulated kurtosis collapsed to 1.6, which indicated that GARCH failed to preserve tail behaviour outside the training window. HMM-NJ achieved 96.5% (SE 0.6) and 96.8% (SE 0.6), and HMM-WJ achieved 95.0% (SE 0.7) and 95.9% (SE 0.6), with simulated kurtosis near 4.9–5.0, confirming that both HMM variants generalised robustly.

Coverage, temporal fidelity, novelty, and diversity. Quantile coverage separated the models: Bootstrap, HMM-NJ, and HMM-WJ maintained 100%, GARCH recovered to 96.0% (from 29.3% IS), and Laplace reached 74.7%, while Gaussian remained low at 36.4%. OoS ACF-MAE values were lower across the board because the shorter window reduced autocorrelation estimation precision: Bootstrap, Gaussian, and Laplace all scored 0.043, GARCH achieved 0.026, HMM-NJ 0.041, and HMM-WJ 0.040. The gap between HMM-NJ and HMM-WJ narrowed from 0.007 IS to 0.001 OoS, which reflected reduced ACF estimation power rather than convergence in temporal quality. Novelty and diversity remained uniformly high across all models (0.948–0.952), confirming that no generator memorised the training data even in a shorter evaluation window.

Nonetheless, the out-of-sample period exposed the stationarity assumption embedded in the IS-calibrated jump parameters. The distribution of KS p -values shifted leftward from IS to OoS, reflecting the expected distributional degradation when stationary parameters were applied to a period with elevated macro uncertainty (Figure 4, panels a–b). The observed ACF of $|G_t|$ in the 2025 window exceeded the HMM-WJ simulation band at medium-to-long lags, indicating that (ϵ^*, λ^*) underestimated the frequency and persistence of volatility episodes in this period and that the

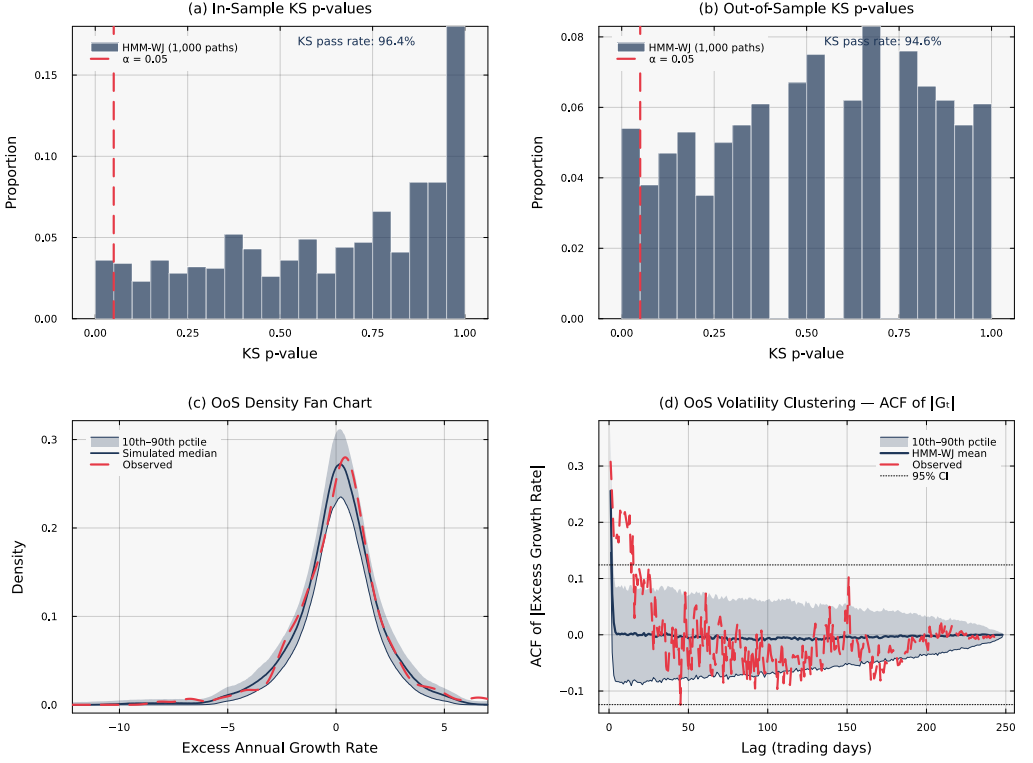


Fig. 4. Statistical validation of HMM-WJ ($N = 100$, 1,000 simulated paths, $\alpha = 0.05$). Panels (a) and (b) show the distribution of two-sample KS p -values in-sample and out-of-sample, respectively; a well-calibrated generative model produces p -values that are approximately uniform above the significance threshold. The leftward shift from (a) to (b) reflects the expected distributional degradation when stationary parameters are applied to a regime with elevated macro uncertainty. Panel (c) shows the out-of-sample marginal density fan chart; the observed density (dashed red) falls within the 10th–90th percentile simulation envelope, confirming adequate distributional coverage despite the regime shift. Panel (d) shows the ACF of $|G_t|$ in the out-of-sample window. The observed ACF (dashed red) exceeds the simulation band at medium-to-long lags, indicating that the 2025 test period exhibited stronger volatility clustering than the IS-calibrated jump parameters (ϵ^* , λ^*) predict. This gap provides direct evidence that the jump hyperparameters are regime-dependent and motivates the time-varying extensions discussed in Section 5.

jump parameters were regime-dependent (Figure 4, panel d). Despite this ACF gap, the observed marginal density remained within the simulation envelope, confirming adequate distributional coverage even under regime shift (Figure 4, panel c).

4.5 Multi-asset extension

We propagated HMM-WJ SPY paths through a Single-Index Model for 424 S&P 500 constituents, with $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$ estimated by OLS on the training window and idiosyncratic shocks drawn by resampling empirical residuals. This yielded a median IS KS pass rate of 66.7% and a mean of 58.4% across the universe (Figure 5, panel b; Table 3). These rates were lower than the 97.0% achieved for SPY alone and reflected the well-known limitation that a single-factor

Table 3. Cross-sectional summary statistics for the Single-Index Model (SIM) extension across S&P 500 constituents. Individual asset paths are generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{SPY,t} + \hat{\eta}_{i,t}$ where $G_{SPY,t}$ is drawn from HMM-WJ ($N = 100$) simulated paths and $\hat{\eta}_{i,t}$ is resampled from empirical residuals. KS pass rates computed over 1,000 paths at $\alpha = 0.05$.

Statistic	Mean	Median	5th pct	95th pct
<i>In-sample (424 assets, 2,766 trading days, 2014–2024)</i>				
KS pass rate (%)	58.4	66.7	2.5	96.0
SIM R^2	0.298	0.295	0.107	0.507
$\hat{\beta}$ (factor loading)	1.066	1.086	0.480	1.657
$\hat{\alpha}$ (intercept)	−0.0328	−0.0196	−0.2037	0.0964
<i>Out-of-sample (417 assets, 249 trading days, 2025)</i>				
KS pass rate (%)	82.1	91.8	23.8	99.0
SIM R^2	0.142	0.142	−0.169	0.479
$\hat{\beta}$ (factor loading)	1.064	1.084	0.482	1.660
$\hat{\alpha}$ (intercept)	−0.0320	−0.0186	−0.2035	0.0965

linear decomposition with symmetric residuals could not fully capture asset-specific tail behaviour, skewness, or sector-level dynamics. The SIM R_i^2 varied widely across GICS sectors, from near-zero for defensive and low-correlation names to above 0.8 for broad-market ETFs (Figure 5, panel a), and the degradation in KS pass rate concentrated among assets with low R_i^2 . High- β assets closely tracking SPY retained pass rates above 80%, which suggested that the HMM-WJ generative engine was not the bottleneck; the limiting factor was the expressiveness of the factor model (Figure 5, panel c).

Out-of-sample KS pass rates across 417 surviving assets reached a mean of 82.1% and a median of 91.8%, confirming that the SIM extension generalised to the test window (Table 3). The correlation-distance distribution between simulated and historical paths yielded a mean distance of $1 - \rho \approx 1.0$, which confirmed that the framework produced temporally independent realisations rather than tracking specific historical events, consistent with its role as a scenario generation tool rather than a forecasting instrument (Figure 4).

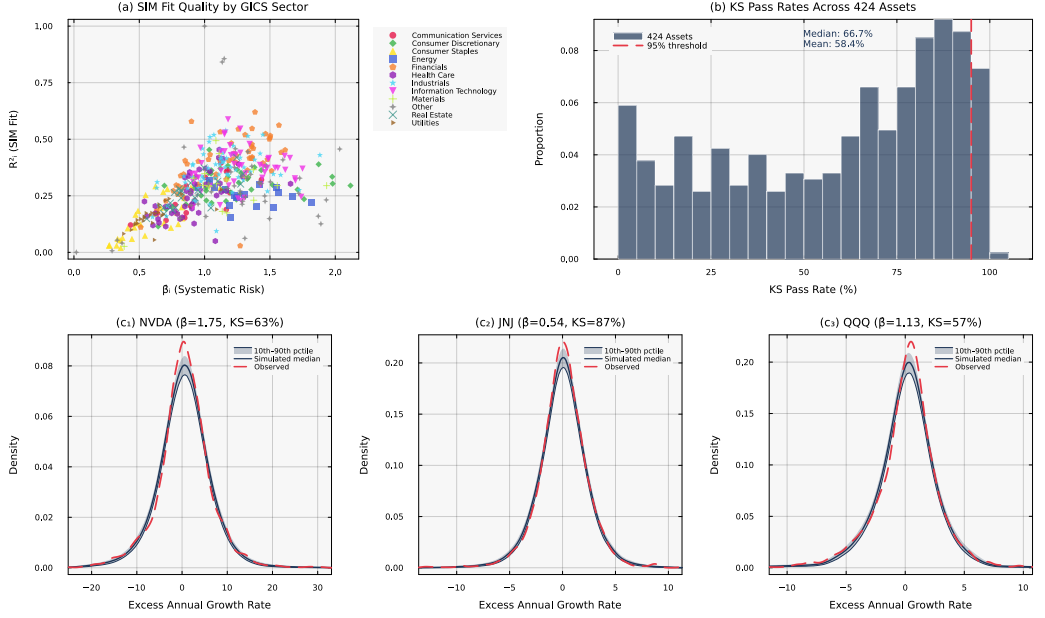


Fig. 5. Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) summarises the SIM regression fit quality by GICS sector. Panel (b) shows that KS distributional consistency is maintained across the full asset universe when paths are generated via the factor structure $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{SPY,t} + \hat{\eta}_{i,t}$. Panel (c) illustrates representative marginal density comparisons for three assets spanning a wide range of systematic risk exposure.

5 Discussion

The empirical results established that the hybrid HMM framework generates synthetic equity paths whose distributional and temporal properties closely match those of historical data, both in-sample and out-of-sample. In this section we interpret the key findings, discuss their practical implications, and identify the limitations that inform the scope of these conclusions.

5.1 The role of the jump-duration mechanism

The comparison between HMM-NJ and HMM-WJ isolated the contribution of the Poisson jump-duration mechanism. Without jumps, the Markov chain transitioned out of tail states within one to two steps on average, producing an ACF of $|G_t|$ that decayed too quickly and failing to reproduce the volatility clustering observed in the data. Introducing jumps forced the model to dwell in high-volatility regimes for empirically realistic durations, and approximately 24% of simulated paths contained at least one such episode, sufficient to shift the ensemble-level ACF into close agreement with the empirical profile. This result is consistent with the broader literature on regime-switching models, where standard HMM specifications have repeatedly been found to undergenerate volatility persistence [9, 37], and it demonstrated that a minimal two-parameter extension (ϵ, λ) could resolve the issue without the computational overhead of hierarchical or semi-Markov alternatives.

The tradeoff between distributional fidelity and temporal structure, visible in the GARCH comparison, deserves emphasis. GARCH(1,1) achieved the lowest ACF-MAE because its variance equation was explicitly parameterised to match autocorrelation dynamics, yet its marginal distribution was systematically inconsistent with the data (IS KS pass rate of 5.5%). The hybrid HMM reversed this

tradeoff: the Laplace-mixture emissions delivered high distributional fidelity, and the jump mechanism provided sufficient temporal structure. Neither model dominated on all metrics simultaneously, but the HMM framework provided a more balanced quality profile for synthetic data applications where both distributional and temporal fidelity matter. Crucially, the volatility-clustering strength of HMM-WJ is not fixed: increasing ϵ raises the fraction of paths that contain jump episodes and drives ACF-MAE toward zero, while decreasing ϵ recovers the pure Markov baseline. The grid search selected the value of ϵ that jointly optimized temporal and distributional fidelity, so the observed ACF-MAE reflects the best achievable balance for SPY rather than a structural limitation of the mechanism.

5.2 Practical implications

The proposed framework offers practical utility for both risk management and the broader synthetic data community. Generative models that reproduce the stylized facts of financial markets enable stress testing under market regimes that are statistically plausible but not historically observed, addressing a key limitation of scenario libraries derived solely from empirical records. The interpretable regime structure, where each hidden state corresponds to a quantile-defined segment of the excess growth rate distribution, facilitates communication between quantitative analysts and risk managers; states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*) and linked to economic narratives. The model's computational efficiency further supports Monte Carlo-based risk metrics such as Value-at-Risk and Conditional Value-at-Risk at scale across large asset universes [17].

The avoidance of the Baum-Welch EM algorithm, made possible by the quantile-based state partition that explicitly assigns each observation to a hidden state, eliminates convergence sensitivity to initialisation and reduces computational cost. This property matters for large-scale pipelines: fitting the model to 424 assets requires only empirical counting and a two-parameter grid search, making it feasible to regenerate synthetic scenarios on a daily or weekly cadence as new data arrive [31, 34]. More broadly, the quality evaluation methodology demonstrated here, combining distributional fidelity tests (KS, AD) with temporal structure metrics (ACF-MAE) and reporting standard errors across Monte Carlo ensembles, provides a reusable template for assessing synthetic time series quality in other domains where temporal dependence structures are critical to downstream applications [4].

5.3 Limitations

Several limitations inform the interpretation of these results. First, the model assumes stationarity of the transition matrix and jump hyperparameters across the full in-sample window. The out-of-sample ACF analysis (Figure 4, panel d) provided direct evidence of this limitation: the IS-calibrated (ϵ^*, λ^*) underestimated volatility clustering persistence in the 2025 test window, consistent with the elevated macro uncertainty of that period. Structural breaks in market microstructure, such as those accompanying regulatory changes or the COVID-19 shock, may not be adequately captured by a single static transition matrix.

Second, the quantile-based state partition is fixed at estimation time and does not adapt to evolving volatility regimes. The boundaries are calibrated on the full in-sample distribution and may overweight the central mass relative to the tails during periods of prolonged market stress.

Third, the Single-Index Model extension imposes a linear, single-factor model of cross-sectional dependence. The median IS KS pass rate of 66.7% across 424 assets confirmed that this decomposition could not capture the heterogeneous tail behaviour, skewness, and sector-specific dynamics present in individual equities [15, 39]. However, the architecture is modular: the HMM-WJ generative engine for the market factor is decoupled from the factor model used to propagate paths, and

replacing the SIM with a richer specification would not require modifying the underlying Markov chain machinery.

Finally, the out-of-sample evaluation covers a single 249-day window (full calendar year 2025), which limits the statistical power of out-of-sample inference. A longer or rolling evaluation would provide a more robust assessment of generalisation performance.

6 Conclusion

We developed a hybrid hidden Markov model that generated synthetic equity paths whose statistical properties closely matched real market data, both in the shape of the return distribution and in the persistence of volatile episodes. The model partitioned excess growth rates into quantile-defined regimes and augmented normal regime-switching with a Poisson jump-duration mechanism that forced the model to linger in extreme states, reproducing the volatility clustering that standard HMMs miss. Applied to SPY over a ten-year training window and validated out-of-sample on the full calendar year 2025, the framework achieved KS and AD pass rates exceeding 95%. The jump mechanism, governed by only two scalar hyperparameters, was the structural feature responsible for generating persistent high-volatility regimes, and the direct frequentist estimation strategy eliminated the computational cost and initialisation sensitivity of the EM algorithm. A Single-Index Model extension scaled the approach to a 424-asset universe, providing a baseline for multi-asset synthetic scenario generation.

Several extensions follow naturally. Time-varying transition matrices, estimated via rolling windows or Bayesian online learning, would allow the model to adapt to structural regime shifts. Replacing the single-factor SIM with multi-factor specifications such as the Fama-French three- or five-factor model [15], principal-component-based factor structures, or nonlinear residual generators could substantially improve asset-level distributional fidelity. Adaptive state definitions based on density clustering rather than fixed quantiles could improve tail coverage during crisis periods. Finally, embedding the hybrid HMM as the generative component within a portfolio optimisation loop, where synthetic scenarios drive tail-risk objectives, would provide an end-to-end application of the framework.

Conflict of Interest Statement

The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions

J.V. directed the study. A.A. developed the model and simulation code, conducted the in-sample and out-of-sample analysis, and generated the figures. Both authors edited and reviewed the final manuscript.

Data Availability Statement

The model code, simulation scripts, training and testing data are available under a Massachusetts Institute of Technology (MIT) license from the GitHub repository: <https://github.com/varnerlab/HMM-w-jumps-paper.git>

References

- [1] Theodore W. Anderson and Donald A. Darling. 1952. Asymptotic theory of certain “goodness of fit” criteria based on stochastic processes. *The Annals of Mathematical Statistics* 23, 2 (1952), 193–212.
- [2] Samuel A. Assefa, Danial Dervovic, Mahmoud Mahfouz, Robert E. Tillman, Prashant Reddy, and Manuela Veloso. 2021. Generating synthetic data in finance: opportunities, challenges and pitfalls. In *Proceedings of the First ACM International Conference on AI in Finance*. 1–8. doi:10.1145/3383455.3422554
- [3] Jay F. K. Au Yeung, Zi-kai Wei, Kit Yan Chan, Henry Y. K. Lau, and Ka-Fai Cedric Yiu. 2020. Jump detection in financial time series using machine learning algorithms. *Soft Computing* 24, 3 (2020), 1789–1801. doi:10.1007/s00500-019-04006-2
- [4] Kevin Bauer, Simon Joussen, and Jan Lukas. 2024. Thinking in Categories: A Survey on Assessing the Quality for Time Series Synthesis. *Journal of Data and Information Quality* (2024). doi:10.1145/3666006
- [5] David Berghaus, Kostadin Cvejovski, Patrick Seifner, César Ojeda, and Ramsés J. Sanchez. 2025. Foundation inference models for Markov jump processes. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*. 129407–129442.
- [6] Jeff A. Bilmes. 1998. A Gentle Tutorial of the EM Algorithm and its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models. *International Computer Science Institute* (1998).
- [7] Fischer Black and Myron Scholes. 1973. The Pricing of Options and Corporate Liabilities. *Journal of Political Economy* 81, 3 (1973), 637–654. doi:10.1086/260062
- [8] Tim Bollerslev. 1986. Generalized Autoregressive Conditional Heteroscedasticity. *Journal of Econometrics* 31, 3 (1986), 307–327. doi:10.1016/0304-4076(86)90063-1
- [9] Jan Bulla and Ingo Bulla. 2006. Stylized facts of financial time series and hidden semi-Markov models. *Computational Statistics & Data Analysis* 51, 4 (2006), 2192–2209. doi:10.1016/j.csda.2006.07.021
- [10] R. Cont. 2001. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance* 1, 2 (2001), 223–236. doi:10.1080/713665670
- [11] José G. Dias, Jeroen K. Vermunt, and Sofia Ramos. 2010. Mixture Hidden Markov Models in Finance Research. In *Advances in Data Analysis, Data Handling and Business Intelligence*. Springer, 451–459. doi:10.1007/978-3-642-01044-6_41
- [12] Francis X. Diebold and Atsushi Inoue. 2001. Long memory and regime switching. *Journal of Econometrics* 105, 1–2 (2001), 131–159. doi:10.1016/S0304-4076(01)00073-2
- [13] Robert F. Engle. 1982. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica* 50, 4 (1982), 987–1007.
- [14] Eugene F. Fama. 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance* 25, 2 (1970), 383–417. doi:10.2307/2325486
- [15] Eugene F. Fama and Kenneth R. French. 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33, 1 (1993), 3–56. doi:10.1016/0304-405X(93)90023-5
- [16] Thomas Fischer and Christopher Krauss. 2018. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research* 270, 2 (2018), 654–669. doi:10.1016/j.ejor.2017.11.054
- [17] Paul Glasserman. 2003. *Monte Carlo Methods in Financial Engineering*. Springer. doi:10.1007/978-0-387-21617-1
- [18] Charles M. Grinstead and J. Laurie Snell. 1997. *Introduction to Probability*. American Mathematical Society.
- [19] James D. Hamilton. 1989. A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica* 57, 2 (1989), 357–384. doi:10.2307/1912559
- [20] James D. Hamilton. 2018. Regime Switching Models. *The New Palgrave Dictionary of Economics* (2018), 11421–11426. doi:10.1057/978-1-349-95189-5_2459
- [21] V. Harwood. 2019. Against the Norm: Modeling Daily Stock Returns with the Laplace Distribution. *arXiv preprint arXiv:1906.10325* (2019).
- [22] Steven L. Heston. 1993. A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options. *The Review of Financial Studies* 6, 2 (1993), 327–343. doi:10.1093/rfs/6.2.327
- [23] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Computation* 9, 8 (1997), 1735–1780. doi:10.1162/neco.1997.9.8.1735
- [24] James Jordon, Lukasz Szpruch, Florimond Houssiau, Luke Sheridan Rains, James Sheridan, Michael Sheridan, and David Sheridan. 2022. Synthetic Data – what, why and how? *arXiv preprint arXiv:2205.03257* (2022).
- [25] Chang-Jin Kim and Charles R. Nelson. 1999. Has the U.S. Economy Become More Stable? A Bayesian Approach Based on a Markov-Switching Model of the Business Cycle. *The Review of Economics and Statistics* 81, 4 (1999), 608–616. doi:10.1162/003465399558472
- [26] A. N. Kolmogorov. 1933. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell’Istituto Italiano degli Attuari* 4 (1933), 83–91.
- [27] Samuel Kotz, Tomasz J. Kozubowski, and Krzysztof Podgórski. 2001. *The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance*. Birkhauser.

- [28] S. G. Kou. 2002. A Jump-Diffusion Model for Option Pricing. *Management Science* 48, 8 (2002), 1086–1101. doi:10.1287/mnsc.48.8.1086.166
- [29] Sohyeon Kwon and Yongjae Lee. 2024. Can GANs Learn the Stylized Facts of Financial Time Series?. In *Proceedings of the 5th ACM International Conference on AI in Finance*. 126–133. doi:10.1145/3677052.3698661
- [30] Benoit Mandelbrot. 1963. The Variation of Certain Speculative Prices. *The Journal of Business* 36, 4 (1963), 394–419.
- [31] Simone Manganello and Robert F. Engle. 2004. Value at risk models in finance. *European Central Bank Working Paper Series* (2004).
- [32] Robert C. Merton. 1976. Option Pricing When Underlying Stock Returns Are Discontinuous. *Journal of Financial Economics* 3, 1-2 (1976), 125–144. doi:10.1016/0304-405X(76)90022-2
- [33] Nguyet Nguyen. 2018. Hidden Markov Model for Stock Trading. *International Journal of Financial Studies* 6, 2 (2018), 36. doi:10.3390/ijfs6020036
- [34] Maureen O’Hara. 1998. *Market Microstructure Theory*. Wiley.
- [35] L. Rabiner and B. Juang. 1986. An introduction to hidden Markov models. *IEEE ASSP Magazine* 3, 1 (1986), 4–16. doi:10.1109/MASSP.1986.1165342
- [36] Alessandro Rossi and Giampiero M. Gallo. 2006. Volatility estimation via hidden Markov models. *Journal of Empirical Finance* 13, 2 (2006), 203–230. doi:10.1016/j.jempfin.2005.09.003
- [37] Tobias Rydén, Timo Teräsvirta, and Stefan Åsbrink. 1998. Stylized facts of daily return series and the hidden Markov model. *Journal of Applied Econometrics* 13, 3 (1998), 217–244. doi:10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V
- [38] G. William Schwert. 1989. Why Does Stock Market Volatility Change Over Time? *The Journal of Finance* 44, 5 (1989), 1115–1153. doi:10.1111/j.1540-6261.1989.tb02647.x
- [39] William F. Sharpe. 1963. A Simplified Model for Portfolio Analysis. *Management Science* 9, 2 (1963), 277–293. doi:10.1287/mnsc.9.2.277
- [40] N. Smirnov. 1948. Table for estimating the goodness of fit of empirical distributions. *The Annals of Mathematical Statistics* 19, 2 (1948), 279–281.
- [41] Elias M. Stein and Jeremy C. Stein. 1991. Stock price distributions with stochastic volatility: an analytic approach. *The Review of Financial Studies* 4, 4 (1991), 727–752.
- [42] Shuntaro Takahashi, Yu Chen, and Kumiko Tanaka-Ishii. 2019. Modeling financial time-series with generative adversarial networks. *Physica A: Statistical Mechanics and its Applications* 527 (2019), 121261. doi:10.1016/j.physa.2019.121261
- [43] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. 2019. Time-series Generative Adversarial Networks. In *Advances in Neural Information Processing Systems*, Vol. 32.

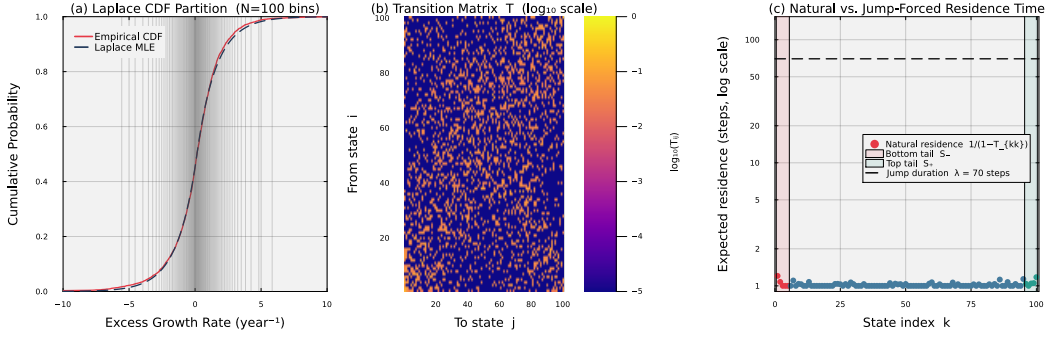


Fig. S1. Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix T in $\log_{10}$ colour scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set S_- (states 1–5, red shading) and top tail set S_+ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean jump duration $\lambda = 70$ steps imposed by the Poisson jump mechanism, quantifying the two-order-of-magnitude gap that motivates the jump-duration extension.

Online Appendix

S1. Fitted model internals

This section visualizes the internal components of the fitted HMM-WJ model for SPY with $N = 100$ states. We constructed the model by fitting a Laplace distribution to the observed excess growth rate series (2014–2024, 2,766 trading days), partitioning the support into N equal-probability quantile bins, and estimating the 100×100 transition matrix from consecutive state assignments (Algorithm 2). Figure S1 displays three diagnostics. Panel (a) confirms that the Laplace marginal closely matches the empirical CDF, validating the distributional assumption underlying the state partition. Panel (b) reveals the banded structure of the transition matrix: most probability mass concentrates near the diagonal, reflecting the tendency of markets to transition between adjacent states rather than making large jumps. Panel (c) exposes the key limitation of the pure Markov dynamics: the natural residence time $1/(1 - T_{kk})$ for every state, including the extreme tails, is only 1–2 steps. This is far too short to reproduce the persistent volatility clustering observed in equity returns, motivating the Poisson jump-duration mechanism described in Section 3.3 of the main text.

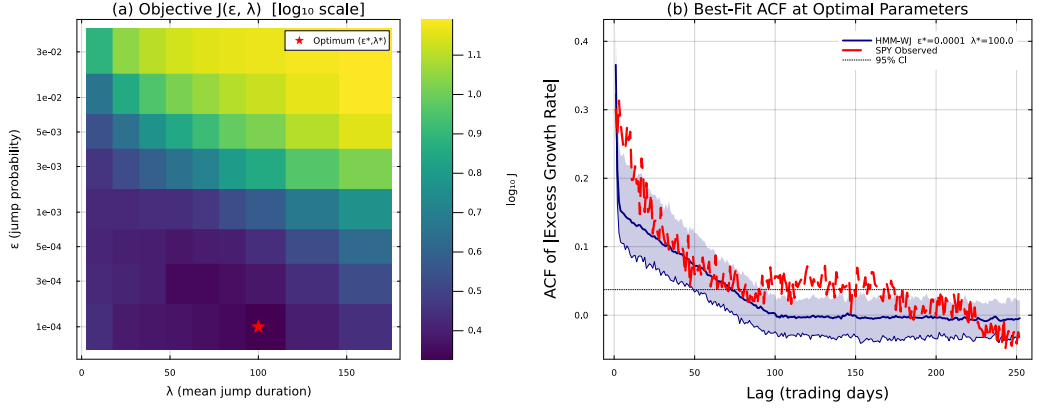


Fig. S2. Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100, 200$ paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid; the red star marks the optimal pair (ϵ^*, λ^*) . The well-defined interior minimum confirms that the jump hyperparameters are identifiable from data. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).

S2. Hyperparameter grid search landscape

The jump mechanism introduces two hyperparameters: the tail-entry probability ϵ and the mean jump duration λ . We selected these via the multi-objective grid search described in Algorithm 4, sweeping $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$ with 200 simulated paths per grid point. The objective $J(\epsilon, \lambda)$ balances two terms: the squared error between observed and simulated absolute-return ACFs (lags 1–252) and a kurtosis penalty weighted by $w_K = 0.20$. Figure S2 presents the results. Panel (a) shows that J has a well-defined interior minimum, confirming that the two hyperparameters are jointly identifiable from data rather than degenerate or weakly constrained. The optimal values $(\epsilon^* = 10^{-4}, \lambda^* = 100.0)$ sit at the boundary of the grid, indicating that the data favor rare but long-lived tail excursions. Panel (b) validates that the ACF of $|G_t|$ at these optimal parameters closely tracks the observed SPY autocorrelation structure, with the simulated 10th–90th percentile envelope (computed from 500 paths) enveloping the empirical curve over the full 252-lag horizon.

Table T1. Sensitivity of model performance to state resolution N . All metrics computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252. Standard errors in parentheses: binomial SE for pass rates, bootstrap SE ($B = 500$) for ACF-MAE.

Model	N	KS pass (%)	AD pass (%)	ACF-MAE
HMM-NJ	200	99.6 (0.2)	98.5 (0.4)	0.059
HMM-NJ	150	99.6 (0.2)	98.4 (0.4)	0.059
HMM-NJ	100	99.0 (0.3)	98.8 (0.3)	0.059
HMM-NJ	90	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	60	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	30	98.9 (0.3)	97.9 (0.5)	0.059
HMM-WJ	200	96.0 (0.6)	88.6 (1.0)	0.049
HMM-WJ	150	96.9 (0.5)	88.8 (1.0)	0.050
HMM-WJ	100	96.8 (0.6)	91.7 (0.9)	0.053
HMM-WJ	90	97.4 (0.5)	91.0 (0.9)	0.052
HMM-WJ	60	96.8 (0.6)	92.4 (0.8)	0.054
HMM-WJ	30	98.0 (0.4)	96.8 (0.6)	0.057

S3. State resolution sensitivity

A natural concern is whether the model’s performance depends critically on the choice of N , the number of discrete states. To investigate this, we re-estimated the full pipeline (Laplace fit, quantile partition, transition matrix, and, for HMM-WJ, the grid search over ϵ and λ) for $N \in \{30, 60, 90, 100, 150, 200\}$ and evaluated each configuration on 1,000 simulated in-sample paths. Table T1 reports KS and AD pass rates at the $\alpha = 0.05$ level along with ACF-MAE (mean absolute error of the absolute-return ACF over lags 1–252). Standard errors are computed via the binomial formula for pass rates and bootstrap resampling ($B = 500$) for ACF-MAE.

For HMM-NJ, distributional fidelity is uniformly high across all resolutions (KS $\geq 98.9\%$, AD $\geq 97.9\%$), confirming that the Laplace quantile partition produces well-calibrated marginals regardless of N . ACF-MAE is effectively constant at 0.059, reflecting the absence of any volatility-clustering mechanism. For HMM-WJ, KS pass rates remain above 96% across all N , while AD pass rates range from 88.6% ($N = 200$) to 96.8% ($N = 30$). The slight decline at high N is expected: finer partitions produce more sparsely populated tail states, making the Anderson-Darling test (which weights tail discrepancies heavily) more sensitive. Crucially, ACF-MAE improves monotonically from 0.057 ($N = 30$) to 0.049 ($N = 200$), demonstrating that the jump mechanism’s ability to reproduce volatility clustering strengthens with finer state resolution. The operating point $N = 100$ balances these two effects, achieving strong performance on all three metrics simultaneously.

Algorithm 2 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f , Number of states N

Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\mathbf{Q}$

- 1: Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t .
 - 2: Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
 - 3: Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
 - 4: Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
 - 5: Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathbf{Q} = \{Q_k\}$.
 - 6: Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros.
 - 7: **for** $t = 1$ to $T - 1$ **do**
 - 8: Identify transition $i = s_t$ to $j = s_{t+1}$.
 - 9: $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
 - 10: **end for**
 - 11: **for** $i = 1$ to N **do**
 - 12: $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ {Row-wise normalization}
 - 13: **end for**
 - 14: **return** $\mathbf{T}, \mathbf{Q}$
-

S4. Additional algorithms

The main text presents Algorithm 1, which generates synthetic state sequences from the fitted HMM with jump-duration dynamics. The three algorithms below detail the remaining stages of the computational pipeline. Algorithm 2 describes model construction: fitting a Laplace distribution to the observed excess growth rates, partitioning the support into N equal-probability bins, assigning each observation to a discrete state, and estimating the transition matrix by row-normalizing the state-to-state count matrix. Algorithm 3 converts a simulated discrete state sequence back into continuous excess growth rates by sampling from a truncated Gaussian within each state's quantile bin, parameterized so that the bin boundaries correspond to $\pm 3.09\sigma$ (the 99.9th percentile). Algorithm 4 performs the multi-objective grid search over the jump hyperparameters (ϵ, λ) , minimizing a composite objective that penalizes both ACF mismatch and kurtosis deviation relative to the observed series.

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Quantile boundaries Q

Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

- 1: Define $z_{0.999} = 3.09$.
 - 2: **for** $k = 1$ to N **do**
 - 3: Set $\mu_k = (Q_{k-1} + Q_k)/2$.
 - 4: Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$.
 - 5: **end for**
 - 6: Initialize the reconstructed excess growth rate vector $\hat{G}$ of length M .
 - 7: **for** $t = 1$ to M **do**
 - 8: Retrieve the simulated state $k = S_t$.
 - 9: Sample the continuous excess growth rate $\hat{G}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$.
 - 10: **end for**
 - 11: **return** $\hat{G}_{1:M}$
-

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$

Require: Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$

Require: Kurtosis penalty weight $w_K = 0.20$

Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

```

1: Calculate observed absolute ACF:  $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$  for  $\tau \in \{1, \dots, L\}$ .
2: Calculate observed global kurtosis:  $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$ .
3: Initialize minimum error  $E_{min} \leftarrow \infty$ .
4: for each  $\epsilon$  in grid space do
5:   for each  $\lambda$  in grid space do
6:     Initialize simulated ACF accumulator  $\mathbf{A}_{sum} \leftarrow \mathbf{0}$  and kurtosis accumulator  $K_{sum} \leftarrow 0$ .
7:     for  $p = 1$  to  $P$  do
8:       Simulate path using Algorithm 1 to obtain state sequence  $S_{1:M}$ .
9:       Decode sequence using Algorithm 3 to obtain continuous rates  $\hat{\mathbf{R}}^{(p)}$ .
10:       $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$  for all  $\tau$ .
11:       $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$ .
12:    end for
13:    Compute ensemble averages:  $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$  and  $\bar{K}_{sim} = K_{sum}/P$ .
14:    Compute objective:  $J = \sum_{\tau=1}^L \left( A_{obs}(\tau) - \bar{A}_{sim}(\tau) \right)^2 + w_K \left( K_{obs} - \bar{K}_{sim} \right)^2$ .
15:    if  $J < E_{min}$  then
16:       $E_{min} \leftarrow J$ 
17:       $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$ 
18:    end if
19:  end for
20: end for
21: return  $\epsilon^*, \lambda^*$ 

```
